Cell

Precise Developmental Gene Expression Arises from Globally Stochastic Transcriptional Activity

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http://dx.doi.org/10.1016/j.cell.2013.07.025

SUMMARY

Early embryonic patterning events are strikingly precise, a fact that appears incompatible with the stochastic gene expression observed across phyla. Using single-molecule mRNA quantification in *Drosophila* embryos, we determine the magnitude of fluctuations in the expression of four critical patterning genes. The accumulation of mRNAs is identical across genes and fluctuates by only ~8% between neighboring nuclei, generating precise protein distributions. In contrast, transcribing loci exhibit an intrinsic noise of ~45% independent of specific promoter-enhancer architecture or fluctuating inputs. Precise transcript distribution in the syncytium is recovered via straightforward spatiotemporal averaging, i.e., accumulation and diffusion of transcripts during nuclear cycles, without regulatory feedback. Common expression characteristics shared between genes suggest that fluctuations in mRNA production are context independent and are a fundamental property of transcription. The findings shed light on how the apparent paradox between stochastic transcription and developmental precision is resolved.

INTRODUCTION

A fundamental question in biology concerns the degree of precision that cellular systems exhibit in their responses to a given set of environmental conditions, extracellular signals, or other input stimuli (Lagha et al., 2012; Lander 2013, Little and Wieschaus, 2011). Production of and interactions between molecules are intrinsically stochastic, limiting the ability of cells to control gene expression and biochemical activities (Raser and O'Shea, 2005), but the propensity of cellular systems to achieve appropriate phenotypic behavior constrains the tolerable magnitude of molecular fluctuations (Rao et al., 2002). In most contexts, it is unknown how closely cellular activity and phenotypic behavior rely on precise control of gene expression.

Many features of *Drosophila* embryogenesis suggest that strict control of gene expression determines reproducible and precise cell fate establishment. In *Drosophila* embryos, patterned gene expression in the early syncytium of ~6,000 nuclei is triggered by asymmetrically distributed, maternally supplied cues (Sauer et al., 1996). Among these is the transcription factor Bicoid (Bcd), the anterior-posterior (AP) concentration gradient of which shows remarkably reproducible distributions between embryos (Gregor et al., 2007). Moreover, within an embryo, the nuclei at similar AP coordinates differ in Bcd concentration by less than 10% (SD over mean), a degree of precision sufficiently high for each row of cells along the AP axis to discern its position from its immediate neighbors (Gregor et al., 2007). Bcd precision correlates with highly precise protein distribution of zygotically expressed target genes (Dubuis et al., 2013; Gregor et al., 2007) that confer cells with distinct gene expression programs within under 3 hr following fertilization (Gergen et al., 1986; Kornberg and Tabata, 1993).

These observations suggest a model in which tightly regulated transcriptional inputs give rise to rapidly established, highly precise outputs. However, the degree of precision in developmental transcription is largely unexplored. In all contexts assayed from prokaryotes to mammalian cells, absolute levels of a given transcript differ by at least ~50% between genotypically identical cells, and for a majority of genes, this variability is even higher (Cohen et al., 2009; Gandhi et al., 2011; Golding et al., 2005; Paré et al., 2009; Taniguchi et al., 2010; Raj et al., 2006, 2010; Reiter et al., 2011; Sigal et al., 2006; Zenklusen et al., 2008). Quantitative observations support the idea that the process of transcription is intrinsically stochastic (Kaern et al., 2005; Li and Xie, 2011). In developmental contexts, it is unknown whether relatively small input transcription factor fluctuations impact the transcriptional output and whether embryogenesis requires the activity of specialized filtering and/or feedback mechanisms to ensure fidelity in the rapid establishment of gene expression programs.

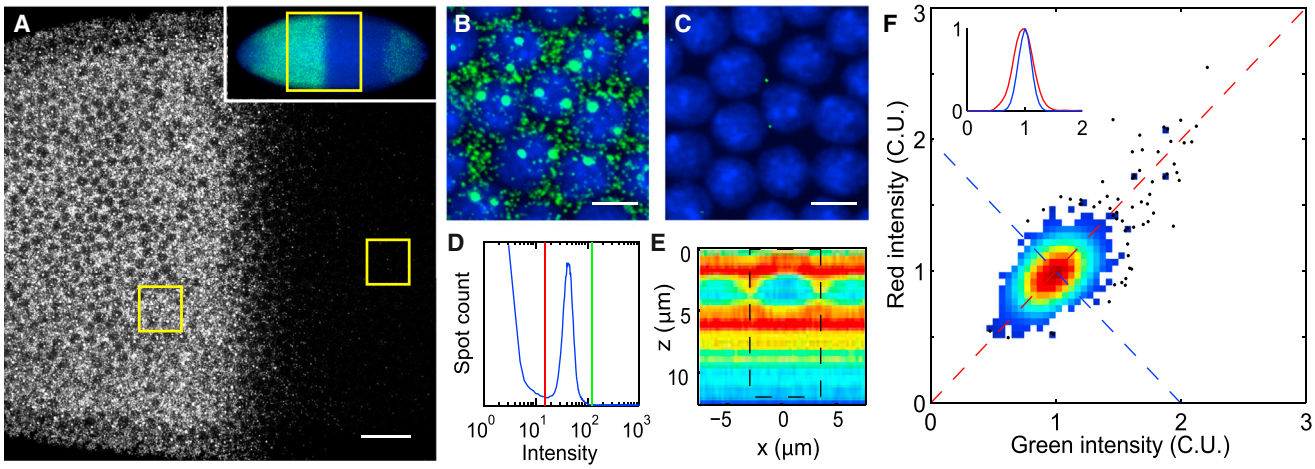


Figure 1. Counting of Absolute Transcript Number in *Drosophila* Embryos

(A) Confocal section through the nuclear layer of a WT embryo during interphase 13 labeled with 114 fluorescent oligonucleotide probes against *hb*, oriented anterior to the left, is shown. Scale bar, 25 μm. Inset is a low-magnification image identifying the region shown in (A).

(B and C) Magnified views of anterior (B) and posterior (C) boxed regions in (A) are presented. hb FISH probes (green) and DAPI staining of DNA (blue). Scale bars, 5 μm.

(D) Particle intensity histogram shows thresholds separating transcripts from noise (red line) and from the long tail of bright transcription sites (green line).

(E) *hb* transcript distribution in axial cross-section through a nucleus centered at x = 0 is shown. z = 0 represents apical surface. Color indicates mean particle density in relative units (red shows high; blue shows low). Dashed box indicates cylindrical summation volume.

(F) Intensity scatterplot in two channels using probes of alternating colors is illustrated. Data point density is given by color; black dots show single point outliers. Inset presents cross-sections of scatterplot in (F) along the correlated (red) and anticorrelated direction (blue) showing Gaussian distributions with σ = 20% (red) and σ = 12% (blue) after normalization to mean cytoplasmic particle intensity (1 "cyto unit" [C.U.]).

See also Figures S1 and S2.

Here, we address these questions with an enhanced method of fluorescence in situ hybridization (FISH) and accompanying image analysis (Little et al., 2011) to label and detect individual zygotically expressed mRNA molecules. We measure in absolute molecular counts the magnitude and fluctuations in the earliest gene expression events of the *Drosophila* embryo. To separate input fluctuations from variability intrinsic to transcription, we focus on those spatial domains in which gene expression is maximally unconstrained. Here, patterning inputs do not determine expression output levels, and thus, input fluctuations cannot impact output variability. These regions thereby reveal the greatest degree of precision achievable by the system. We show that in these regions, the earliest expressed genes share common expression characteristics: despite their expression in spatially distinct territories, their rates of production are identical, and all display intrinsically stochastic transcriptional activity. These similarities suggest that expression rate and variability result from fundamental, global features of transcriptional regulation that limit the attainable degree of precision. Nevertheless, the stochastic expression results in precise and nearly uniform transcript accumulation, achieved by straightforward spatiotemporal averaging.

RESULTS

Measuring Absolute Numbers of mRNA Transcripts in Early *Drosophila* Embryos

Previous work in *Drosophila* embryos has documented that nuclei at similar AP coordinates express nearly equivalent protein amounts of the gap gene Hunchback (Hb) with fluctuations of <10% (Gregor et al., 2007). The transcriptional activator of Hb, Bcd, displays variability on the same order as Hb (Gregor et al., 2007). A precise transcriptional response of the *hb* locus presents the most straightforward though as yet untested explanation of minimal Hb variation. To quantitatively evaluate transcription of *hb*, we adapted a FISH method developed previously (Little et al., 2011) to label *hb* mRNAs using multiple fluorescently labeled antisense DNA oligonucleotides (Figure 1A). By scanning confocal microscopy, we detect two broad classes of objects: sparse, bright spots representing sites of nascent transcription (e.g., Wilkie et al., 1999); and numerous diffraction limited spots, ~90% of which are located in the internuclear space that we refer to as cytoplasmic particles (Figures 1A–1C). These particles exhibit sufficiently high contrast to be readily distinguished from background imaging noise using automated image processing (Figure 1D). Each particle is found on at least three adjacent 250 nm confocal imaging sections with three-dimensional structure identical to the measured point spread function (PSF; Figures S1A–S1D available online). To test detection efficiency, we applied probes with alternating fluorophore colors. A minimum of 85% of cytoplasmic particles detected in one channel are found in the other, indicating that >94% of mRNAs are detected in at least one channel (Figures S1E–S1G).

Tight unimodal clustering around mean intensity suggests that the cytoplasmic particles are similar in mRNA content (Figure 1D). Deviation from mean intensity results from at least two phenomena: particles can be bound by different probe numbers, and multiple particles can overlap and be detected as single

spots. To determine the relative contributions of each, we examined correlation of intensities in two-color detection. Correlation is weak (Figure 1F), implying that the fractional SD of mRNA content in detected particles is at most 16% (see Experimental Procedures). To determine the number of mRNAs per particle, we compared counts of maternally deposited *hb* mRNA particles in entire imaged embryos to those from quantitative RT-PCR (Figure S1H). We found an average of 1.2 ± 0.5 mRNAs per imaged particle. The low mean indicates that the probability of detecting more than two mRNAs per particle is essentially zero. This observation, coupled with the SD in mRNA content from imaging, determines a probability of 97% of finding one mRNA per particle (see Experimental Procedures). Moreover, comparing counts of wild-type (WT) and *hb* hemizygous embryos yields a 2-fold concentration difference (Figure S1I), lending further support to the validity of our approach.

Because the density of zygotically produced *hb* mRNA increases above about one molecule per μm^3, PSFs of individual molecules begin to overlap. We extend particle counting to arbitrarily high density using the naturally large dynamic expression range. We determine counts in dense regions by measuring total fluorescence collected from all mRNAs per volume and calibrating to low-expressing regions where individual mRNAs are counted directly (Figure S2A). We thereby measure absolute concentration and local fluctuation with accuracies of 12% and 5%, respectively (Figures S2B and S2C). The two methods have overlapping domains of applicability: direct counts are accurate for transcript concentrations $\lesssim$0.5 molecules/μm^3, and total fluorescence for concentrations $\gtrsim$0.35 molecules/μm^3 (Figure S2C). Thus, our FISH method is suitable for high-precision measurements of absolute mRNA counts at any density.

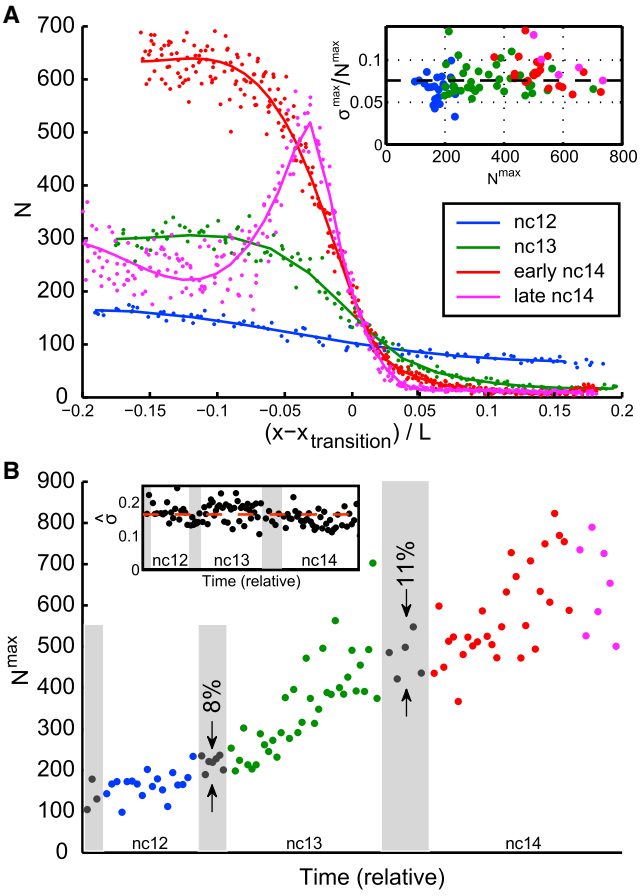


Figure 2. Precision and Reproducibility of Cytoplasmic *hb* Profiles

(A) Absolute cytoplasmic *hb* mRNA counts per standardized volume as a function of AP position are presented. Data for four embryos at nc12 (blue), 13 (green), early 14 (red), and late 14 (magenta) are illustrated. Position is shown as distance from inflection point $x_{transition}$ (see also Figure S3C). Inset presents fractional SD σ^{max}/N^{max} in the spatial domain of highest mRNA accumulation as a function of the mean count (N^{max}) for 101 embryos. Dashed line is at 8%.

(B) Cytoplasmic *hb* mRNA counts (N^{max}) as a function of time is shown. Ages estimated by visual inspection of DAPI staining are illustrated; relative width of mitoses (gray shading) and interphases according to Foe and Alberts (1983) is presented. Reproducibility of counts in 12th and 13th mitoses is 8% and 11%, respectively. Inset shows the estimated reproducibility $\hat{\sigma}$ as a function of time. Data points indicate running averages of root-mean-square displacement from smoothed timeline over 15 consecutive data points normalized to mean. Dashed line represents average $\hat{\sigma}$ (17%).

See also Figure S3.

Cytoplasmic *hb* mRNA and Protein Distributions Display Similar Levels of Precision

To assess fluctuations in transcript number between nuclei, we measure mRNA concentration in cylinders separated by one internuclear distance to a depth of 12 μm beneath the plasma membrane, encompassing the majority of zygotically expressed transcripts (Figures 1E and S3A). As expected from prior observations (e.g., Tautz et al., 1987), *hb* transcripts accumulate dramatically in the embryo anterior during early blastoderm (Figure 2A). Transcription is terminated early in the interphase of nuclear cycle 14 (nc14) except near the embryo midpoint, and maternally supplied transcripts are continuously lost from the posterior (Figures 2A, S3B, and S3C). *hb* mRNA expression profiles correlate well with observed protein levels (Figure S3D).

As an initial quantification of precision, we ascertain the degree of variability independent of putative regulatory inputs by examining the spatial domain of maximal transcript accumulation, i.e., nuclei found in regions of highest-observed gene product levels. Here, expression noise (the fractional SD of *hb* concentration) is 8% ± 2% as early as nc12; thus, *hb* mRNA levels exhibit equal or better precision than Hb protein (Gregor et al., 2007). Age-ordered embryos show a monotonically increasing count through mid-nc14 (Figure 2B), with an approximately constant fractional SD across embryos (17% ± 3%, Figure 2B inset). Ambiguous age determination in fixed samples results in large fluctuations across embryos of approximately the same age. This effect is minimized in embryos undergoing mitosis, during which transcription ceases (Shermoen and O'Farrell, 1991), allowing unambiguously temporal ordering. Counts differed by less than 11% in mitotic embryos, similar to the degree of reproducibility in Hb protein profiles (Gregor et al., 2007). The actual precision and reproducibility are likely to be higher because our measurements contain systematic errors arising from the FISH procedure such as physical distortion (5% measurement error) and error in counts (2%–3% measurement error; see Experimental Procedures and Figure S2C).

Importantly, variation of cytoplasmic profiles is nearly at the level of Poisson counting noise, i.e., at the lowest bound that can be attained by a stochastic process. For 500 molecules per volume, as observed in late nc13 or early nc14, counting noise amounts to 5%, matching the lower bound of our measurements (Figure 2A, inset). Large mRNA counts provide a natural buffer against potential fluctuations in translation that have been observed in other systems (Bar-Even et al., 2006; Newman et al., 2006; Taniguchi et al., 2010), yielding precise Hb expression. By comparison, in genome-wide studies, the most highly (and therefore most precisely) expressed genes in yeast and *E. coli* exhibit cell-to-cell fluctuations exceeding 50% in mRNA count (Gandhi et al., 2011; Taniguchi et al., 2010). Thus, early embryos exhibit an extraordinary degree of precision, rarely observed in other contexts.

Our timeline suggests that *hb* transcript lifetime is large because we see no decrease in counts during the 12th and 13th mitoses (Figure 2B). We verified this by measuring *hb* mRNA lifetime directly, disrupting transcription with α-amanitin injection and subsequently monitoring loss of zygotic *hb* (Figures S3E and S3F). We find a lifetime of $\sim$60 min, consistent with an estimate from imaging: transcript loss of <11% (Figure 2B) in 5 min of mitosis (Foe and Alberts, 1983) corresponds to a lifetime of >45 min. These results show that the accumulation of transcripts is only mildly impacted by degradation.

Determining Instantaneous Transcriptional Activity by Measuring Total Nuclear Nascent mRNA Content

The low noise of *hb* cytoplasmic mRNA counts suggests that nuclei in the fully active region produce transcripts at nearly equivalent rates. However, all systems studied to date, including *E. coli*, yeast, cultured cells, and late *Drosophila* embryos (Golding et al., 2005; Larson et al., 2011; Paré et al., 2009; Raj et al., 2006; Zenklusen et al., 2008), produce transcripts through brief intervals of dense output interspersed with long quiescent periods of stochastic duration (Li and Xie, 2011). This seems incompatible with near uniformity of cytoplasmic mRNA content. To determine the extent of variability in transcriptional activity, we developed a measure of transcription using the fluorescence intensities of nascent transcription sites.

Consistent with previous results from Wilkie et al. (1999), we observe that the maximum number of detectable nascent sites per nucleus increases from DNA replication during interphase from two sites early to four at mid-to-late interphase (Figure S4A). Because sister chromatid loci remain in close physical proximity until mitosis, and because transcription sites occasionally occupy overlapping focal volumes, the number of active loci is challenging to discern. Instead, we used the total fluorescence of all transcription sites in a nucleus as a measure of instantaneous transcriptional activity. Assuming that nascent and mature mRNAs are equally accessible to probes, nascent site intensities can be represented as an equivalent number of mature cytoplasmic mRNAs by normalizing to the mean or ''unit'' intensity of completed transcripts, yielding transcriptional activity in absolute units of total mRNA content. To determine the extent of measurement error arising from differences in probe binding affinity and/or the subsequent normalization procedure, we used probes of alternating fluorophore colors. Ideally, for a given nascent site, the number of cytoplasmic units (C.U.) will be identical in both colors. Plotting nascent mRNA content of one color as a function of the other yields points on a line with a slope close to unity (among five embryos, the mean slope [±SD] is 0.90 ± 0.09), with a scatter of 5% (Figure 3A). We thus measure transcriptional activity with an error of 5% and relate it to absolute mRNA content with an uncertainty under 20% (the largest deviation of 0.90 ± 0.09 from 1).

Three lines of evidence support the idea that nascent mRNA content reflects instantaneous transcriptional activity. First, the appearance of loci is coupled to the nuclear mitotic cycle: they are observed during interphase and absent during mitosis. Second, transcription in anterior nuclei initiates slightly earlier than in those closer to the center of the embryo, both because anterior nuclei inhabit a region of higher concentration of Bcd and because of metasynchronous nuclear divisions propagating as a wave toward the embryo center (Foe and Alberts, 1983). Consistent with expectation, during the first minute of the 13th interphase, we observe a gradient of nascent mRNA content along the AP axis (Figure S4B). Third, we designed probe sets to label the 5′ and 3′ portions of the completed transcript with fluorophores of green and red colors, respectively. If nascent sites are composed of incomplete transcripts, then 5′ sequences must be more numerous than 3′ sequences (Figure S4C), resulting in an increase of green signal at the expense of red. In agreement, our measurements reveal the enrichment of green signal after normalization (Figure 3B). Importantly, greater 5′ enrichment is observed as the fraction of transcript labeled with green fluorophore increases (increasing slopes of fit lines in Figure 3B). Thus, nascent *hb* loci are largely composed of unfinished transcripts and serve as a measure of transcriptional activity.

Variation in Nascent Transcription Site Activity Is 6-Fold Higher than Variation in Cytoplasmic Output

Given the low noise in cytoplasmic counts, we expected that the nascent mRNA content at all genomic loci would rise simultaneously until saturated with RNA polymerase II (RNAP), in principle reaching and sustaining some maximum nascent mRNA content. However, our measurements of nascent mRNA content show otherwise (Figures 3C–3E). The nuclear nascent mRNA content varies by 22% ± 3%, 3-fold higher than that observed in cytoplasmic counts. To be certain that this variability does not result from the delay in attaining steady-state maximum activity after mitosis (Figure S4D), we confined our analysis to mid and late interphase 13 embryos. We observe this degree of variation even when loci are allowed the full temporal extent of interphase 13 to reach a putative maximum (Figure 3E). These results indicate that *hb* loci fail to sustain any amount of uniform maximum content.

The 22% variation we observe reflects fluctuations across a maximum of four active genomic loci in each nucleus of WT embryos. If transcription from each locus acts independently, then the variability between nuclei must decrease in proportion to the root of the number of loci; thus, the expected variability between individual loci is $22\% \times \sqrt{4} = 44\%$. To test whether loci are in fact independent, we examined embryos heterozygous for a *hb* deficiency in which each nucleus possesses a maximum of only two loci. We observed that total nascent mRNA content

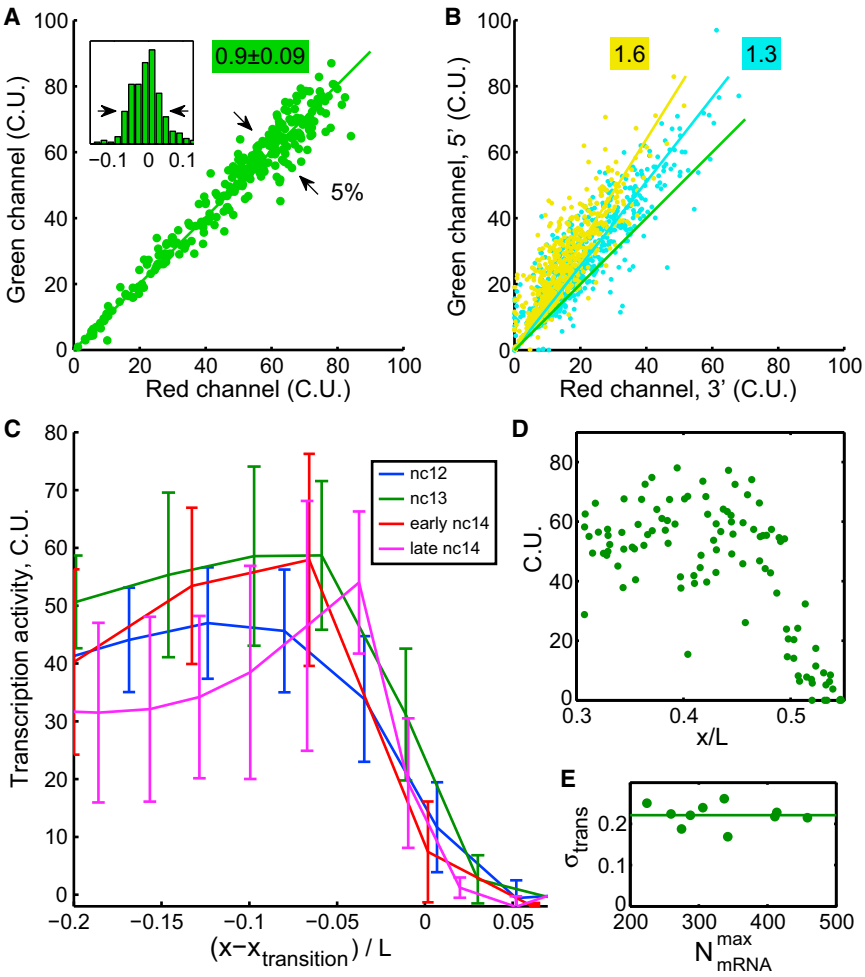


Figure 3. Variability of Transcriptional Activity at Nascent Transcription Sites

(A) Scatterplot of total nascent *hb* mRNA per nucleus using probes of alternating colors for an embryo in nc13 after normalization to the mean cytoplasmic particle intensity (C.U.) is illustrated. Intensities follow a direct proportionality relation with slope 0.90 ± 0.09 (n = 5 embryos). Inset shows root-mean-square normalized deviation from linear fit; scatter = 5% (arrows).

(B) Two-color scatterplot of nascent mRNA content in which probes bearing the same fluorophore are clustered on the 5′ (green channel) and 3′ (red channel) portions of the transcript is shown. Cyan indicates measurements using 57 green and 57 red-labeled probes; observed slope is 1.3. Yellow represents results with 78 green and 36 red-labeled probes; observed slope is 1.6. Green line in (A) is plotted for comparison.

(C) Transcriptional activity per nucleus as a function of position along the AP axis for four embryos in nc12 (blue), 13 (green), 14 early (red), and 14 late (magenta) in binned averages of 10, 20, 40, and 40 nuclei, respectively, is shown. Error bars, SDs within bins. Position is shown as distance from inflection point $x_{transition}$.

(D) Transcriptional activity per nucleus as a function of absolute AP position for the embryo in interphase 13 in (C) is presented.

(E) Transcription noise for ten embryos in nc13 is plotted as fractional SD across nuclei versus cytoplasmic *hb* counts within the spatial domain of highest accumulation. Transcription activity noise remains constant throughout interphase at 22% ± 3%.

See also Figure S4.

per nucleus varies by 33% ± 6%, which corresponds to a transcriptional activity of $33\% \times \sqrt{2} = 47\%$ in individual loci. This number is nearly identical to WT and, hence, consistent with independence. The variability between individual loci of ~45% represents a 6-fold increase over fluctuations in cytoplasmic counts. Analyzing closely apposed alleles on sister chromatids with sufficient separation to reliably gauge intensities reveals no correlation in their activities (Figure 4A), indicating independent activity even for recently duplicated loci. If the observed fluctuations result from variability in any input factor controlling *hb* expression (i.e., "extrinsic" noise), then the variation in total nuclear activity would show no dependence on the number of loci in the nucleus. However, because the noise scales with the number of loci (Figure 4B), the fluctuations we observe in the maximally expressed domain are intrinsic to the process(es) of transcription and not determined by variability in the controlling inputs.

Transcriptional activity will necessarily exhibit some degree of noise arising from stochastic single-molecule events, but the fluctuations we observe exceed the Poisson expectation considerably. From our observations, we can estimate the number of RNAPs engaged in transcription per nucleus and thereby determine the expected degree of fluctuations; we find the predicted noise magnitude of at most 11% (see Experimental Procedures). The observed fluctuations of 22% ± 3% are at least 2-fold greater than this prediction, ruling out a model in which transcriptional fluctuations in the region of maximum expression are determined by a single rate-limiting step of RNAP loading (Figure S4C).

From these observations, we conclude that first, even in the domain of maximal expression, *hb* is not saturated with the maximum possible density of RNAP, and second, despite the near uniformity of cytoplasmic transcript concentration, instantaneous activity of individual *hb* loci is intrinsically stochastic. The estimated variation in transcriptional activity at an individual locus is very similar to the minimum value of ~50% observed for differences in mRNA numbers for the most highly expressed genes in yeast and *E. coli* (Gandhi et al., 2011; Taniguchi et al., 2010). In all contexts, variation between cells is significantly higher than that predicted for a process with a single rate-limiting step (Chubb et al., 2006; Golding et al., 2005; Le et al., 2005; Raj et al., 2006). These similarities across such diverse contexts suggest that the observed fluctuations are globally inherent features of the activity of otherwise "fully activated" genes. In the context of a rapidly developing embryo, the highest-attainable

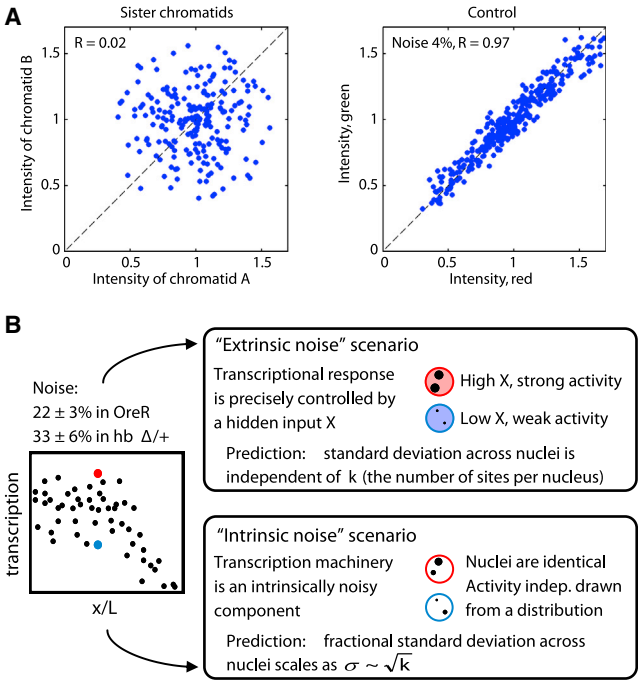


Figure 4. Fluctuations in *hb* Transcription Are Dominated by Intrinsic Noise

(A) Transcriptional activity of loci on optically resolved sister chromatids is uncorrelated (Pearson correlation coefficient R = 0.02), compared to the tight correlation (R = 0.97) in a control experiment using probes of alternating colors (with 4% imaging noise).

(B) Transcriptional variability arises from fluctuations in inputs (extrinsic noise) and from the process of transcription itself (intrinsic noise). Two extreme scenarios are presented in cartoon form. Upper: a fluctuating extrinsic input leads to correlated activities of transcription sites within a given nucleus; its contribution to the fractional SD is independent of the number of transcribing loci k. Lower: intrinsic mechanistic noise affects all transcription sites independently; the fractional SD scales as the inverse square root of available transcription sites. Left view shows the measured transcription noise in WT and *hb*Δ/+ embryos (22% ± 3% and 33% ± 6%, respectively) showing scaling behavior characteristic of intrinsic noise with magnitude ~45% ($\sqrt{4} * 22\% = 44\%$; $\sqrt{2} * 33\% = 47\%$).

expression rate would serve to minimize cell-to-cell fluctuations to the fullest possible extent and thereby promote precision. The observed tolerance of fluctuations, linked with the apparent inability to sustain saturating RNAP density, suggests that this degree of imprecision cannot be circumvented even in this highly precise developmental context.

The Magnitude of Expression Noise Is Independent of Autoregulation and Transcriptional Modulation

Our results suggest that in addition to fluctuations from controlling inputs, *hb* activity possesses a large inherently stochastic component. We examined whether the inherent noise could be attributed to features of *hb* regulation. First, the *hb* locus contains several binding elements for Hb protein itself, and genetic evidence indicates that Hb is required for its own expression (Holloway et al., 2011; Margolis et al., 1995; Treisman and Desplan, 1989). Positive feedback will amplify fluctuations if locally produced mRNA and protein dominate autoregulation (i.e., if diffusion is limited). Second, noise necessarily decreases as RNAPs approach their maximum loading density along the gene, but transcriptional repressors might disallow high densities, resulting in greater variability.

To determine the effect of positive feedback, we examined mRNA production in embryos homozygous for an early *hb* stop codon. Mutants and WT siblings display similar expression patterns until mid-nc14 when WT embryos show reduced anterior expression and the loss of accumulated transcripts (Figures 5A and 5B). In mutants, *hb* is maintained, resulting in continued high transcript density at late times (Figure 5A; Margolis et al., 1995). However, the absence of autoinhibition did not alter the magnitude of variation in transcriptional activity compared to WT siblings, supporting the idea of intrinsic fluctuations and suggesting that zygotic Hb inhibits its own expression (Li et al., 2008; Perry et al., 2012; Treisman and Desplan, 1989).

We also examined whether the removal of transcriptional inhibitors would allow the accumulation of larger numbers of RNAP. First, we examined *hb* expression in *runt* homozygous mutants because Runt is implicated in gap gene regulation (Chen et al., 2012; Li et al., 2008;Tsai and Gergen, 1994). *hb* expression differs from WT starting at mid-nc14 when *runt* embryos maintain high levels of *hb* transcript in the anterior 20%–40% (Figure 5C). *hb* profiles in *runt* mutants tend to resemble younger WT siblings, reaching WT levels near the end of nc14 (Tsai and Gergen, 1994), suggesting a delay in the reduction of anterior expression. In support, we find that *runt* mutants express slightly greater transcriptional activity than WT embryos with similar expression profiles (Figure 5D). However, at no time does expression noise differ noticeably from WT. Therefore, *runt* activity is required for the correct timing of *hb* downregulation but plays no discernable role in determining fluctuations in transcriptional activity.

Finally, we tested the effect of manipulating the activity of maternally provided transcriptional modulators. We impaired the activity of the repressors Capicua and the corepressor Groucho by mutation and observed that although *hb* expression boundaries were altered consistent with previous observations (e.g., Margolis et al., 1995), *hb* expression noise in the anterior was not affected. We also altered the genetic dosage of *bcd* to provide between 50% and 280% of WT activity, which shifted *hb* expression along the AP axis as expected (Liu et al., 2013) but had no effect on expression variability (Figures S5A and S5B). Thus, our data are consistent with the view that the fluctuations in *hb* transcriptional activity arise from intrinsically stochastic processes, independent of variability in transcriptional modulators.

Gap Genes Share Expression Characteristics and Are Produced at Equal Rates

hb transcription fluctuates around a mean polymerase density that is about half the level that is physically obtainable. What sets the magnitude of the mean activity? Specific features of the *hb* promoter may limit activity. Alternatively, the maximum rate may not be specific to *hb* but shared among early expressed genes. If so, this would suggest that the maximum obtainable output, and its related noise level, is not set by any specific

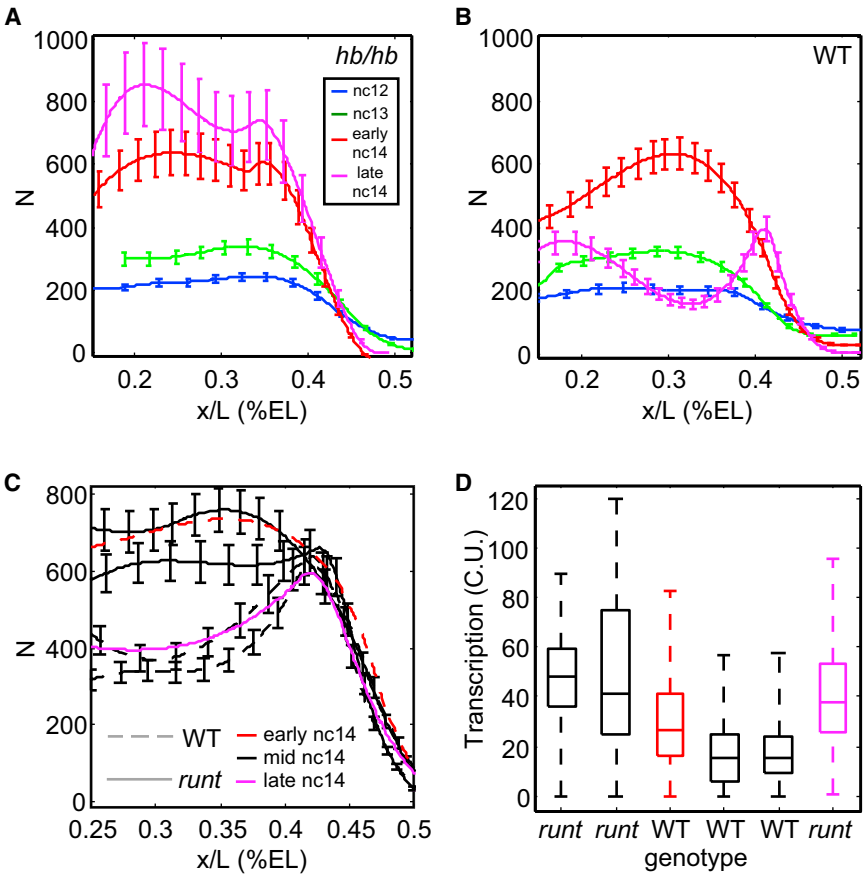


Figure 5. Mutations in *hb* and *runt* Impair Timely Repression of *hb* Expression

(A and B) Cytoplasmic *hb* mRNA counts per standardized volume as a function of AP position of similarly staged zygotic *hb* mutants (A) and WT siblings (B) are presented. Smooth profiles are best spline fits. Error bars are root-mean-square deviations from smoothened profiles, calculated in windows of ten nuclei.

(C and D) *runt* mutants show delayed repression of *hb* transcription in nc14.

(C) Solid lines indicate *runt* mutants; dashed lines show WT siblings. Black represents embryos of similar age (mid nc14) as judged by DAPI. Profile of an early nc14 WT embryo (red line) resembles midstage *runt* mutants, whereas a very late *runt* mutant (magenta line) is similar to earlier WT siblings.

(D) Transcriptional activity in anterior nuclei of the embryos shown in (C) is presented. Box plot depicts median, quartiles, and range of nascent transcription site activity for each embryo. *runt* mutants display consistently higher *hb* activity compared to WT siblings, leading to the inappropriately high *hb* transcript counts at late times as shown in (C).

See also Figure S5.

promoter-enhancer arrangement or any patterning cue and, instead, is determined by general physical considerations.

To discern between these possibilities, we measured the accumulation of transcripts of four gap genes primarily responsible for trunk patterning: *hb*, *Krüppel* (*Kr*), *knirps* (*kni*), and *giant* (*gt*) (Figure 6A). We found that all four genes display nearly uniform accumulation of cytoplasmic mRNAs accompanied by over 3-fold higher fluctuations in instantaneous transcriptional activity, essentially identical to the characteristics of *hb* (Figures 6B and 6C). These results strongly suggest that all early transcriptional events are subject to common constraints.

To closely compare transcript accumulation between genes, we took advantage of the observation that for *Kr* and *kni*, cytoplasmic mRNA density increases monotonically between early nc12 and well into interphase 14 (Figures S5C and S5D), in contrast to *hb*, which ceases accumulating broadly in early nc14 (Figure 2A). We used counts of *Kr* or *kni* as a proxy for time, reducing staging uncertainty when comparing different genes. We performed dual-color labeling with probes against pairs of gap genes and report cytoplasmic counts of *hb*, *gt*, and *kni* mRNA as a function of *Kr* (Figure 6).

Figure 6C displays the expression of *hb* and *Kr* in WT (blue), *hb* hemizygous (green), and Kr^1 heterozygous (red) embryos during nc12 and nc13. Counts in deficiency or mutant heterozygous embryos deviate considerably from WT (Figure 6C, inset), but after multiplying the counts of the deficient gene by 2, all points collapse onto the same line (Figure 6C), showing the absence of compensatory mechanisms. We observed the same behavior for *gt-Kr* and *kni-Kr* expression pairs. Unexpectedly, for the three sets of gene pairs, linear fitting yields lines with slopes between 0.9 and 1.15 (Figure 6E); that is, in their regions of maximal expression, the four genes are produced at nearly identical rates. The differences between absolute levels within these regions (Figure 6E) reflect differences in the timing of when *kni*, *Kr*, and *gt* transcripts begin to accumulate, and for *hb* the perdurance of maternal mRNA. These maximal production rates are independent of Bcd activator concentration: although genetically altering the dosage of *bcd* between 50% and 280% of WT shifts the expression domains along the AP axis (Liu et al., 2013), this manipulation does not alter either accumulation rate or precision (Figures S5A and S5B).

These results are consistent with the idea that these transcripts are produced at the same rate. This strong similarity occurs despite the fact that these genes are expressed maximally in nonoverlapping spatial domains. The transcriptional activity in the maximally expressed domain and the magnitude of transcriptional noise are therefore set independently of the inputs that determine spatially patterned expression, which are specific to each gene. By focusing on the regions of maximal expression, we could isolate the features of transcription that appear to be universal across gap genes and, furthermore, match the noise characteristics previously observed in bacteria and cell cultures. This suggests that the failure to sustain a maximal loading of RNAPs on

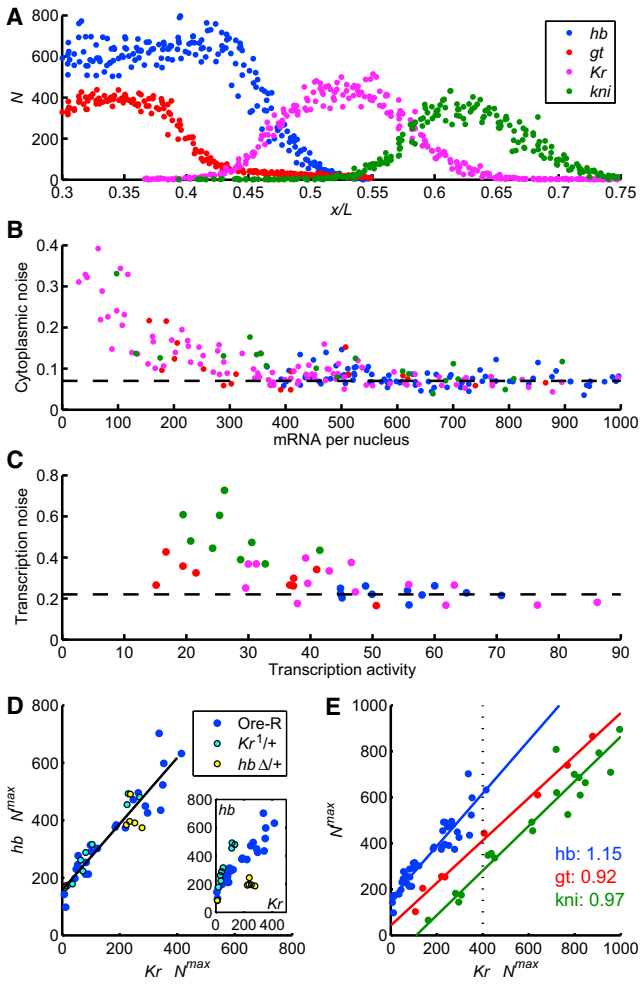


Figure 6. Universal Properties of Gap Gene Transcription

(A) Cytoplasmic profiles are shown for four gap genes (mRNA concentration per standard volume as a function of AP position) measured in two embryos of the same age (second half of nc13; indicated by dotted line in E) processed with *hb* (blue) and *gt* (red) and with *Kr* (magenta) and *kni* (green) probes.

(B–E) Gap gene expression characteristics within each gene's region of maximum expression are presented.

(B) Noise in cytoplasmic counts as a function of counts per nucleus (dashed line indicates 8%) is illustrated.

(C) Noise in transcriptional activity as a function of activity level (dashed line indicates 23%) is shown.

(D) mRNA expression (mean mRNA count per standard volume) in embryos from cycle 12 to early 14 costained with FISH probes against *hb* and *Kr* mRNA is illustrated. Data from WT embryos (blue) coincides with those from embryos deficient for one copy of *hb* (*hb*Δ/+; yellow) or *Kr* (Kr^1/+; cyan) when the concentration of the respective mRNA is rescaled by a factor of 2. Inset shows raw data (not rescaled).

(E) Levels of *hb* (blue), *kni* (green), and *gt* (red) versus *Kr* are shown. Data from WT, *hb*Δ/+, and Kr^1/+ embryos are combined by rescaling as in (D) (also see Figures S5C and S5D). *hb* data as in (A) are presented; *kni* and *gt* were assessed in cycles 13 and 14. Slopes of fit lines indicate ratio of absolute production rates; all are within 15% of unity.

the gene and the intrinsic noise of 45% are a common feature of transcriptional activation across diverse biological contexts.

DISCUSSION

The fundamental question of how embryos achieve precise control over the earliest transcriptional events is largely unanswered. General models of embryogenesis posit that early developmental events are dominated by molecular noise and imprecision in the control of gene expression (Arias and Hayward, 2006; Manu et al., 2009; Rao et al., 2002), a view consistent with observations of wide fluctuations in transcriptional activity in the majority of systems assayed quantitatively (Li and Xie, 2011). Indeed, the finding in fly embryos that instantaneous transcriptional activity varies between loci by nearly 50% suggests that early transcription in fly embryos obeys rules of stochastic activity observed in other systems where output can vary by a similar degree and is often much higher (Munsky et al., 2012). Stochastic variation appears to be a universal feature of transcription from single-cell organisms grown in culture (Raj et al., 2006; So et al., 2011; Stewart-Ornstein et al., 2012) and for cells in certain developmental settings (Paré et al., 2009; Raj et al., 2010; Saffer et al., 2011). *Drosophila* embryos display an extraordinary degree of precision in the rapid establishment of distinct gene expression programs; nevertheless, even this system cannot circumvent stochastic transcriptional activity. This finding supports the idea that control systems that might overcome stochastic molecular activity are difficult to design, costly to implement, and rarely if ever found (Lestas et al., 2010).

It is possible that cultured yeast and bacteria exist at sufficiently high densities such that precision is not required to ensure survival of a large fraction of the population; indeed, in several cases, stochastic expression serves to maximize survival options (Balaban et al., 2004; Maamar et al., 2007; Mirouze et al., 2011; Nachman et al., 2007). In addition, for prokaryotes and haploid yeast and unlike early embryos, the presence of a single genomic locus precludes the possibility of noise filtering by averaging over independent loci. Alternatively, precision might be required to ensure the survival of single-cell organisms when grown in their endogenous conditions, which may be difficult to study in a laboratory setting. However, in stark contrast to single-cell organisms, many developing embryos possess large fields of cells that must coordinately undertake rapidly determined fate decisions, thus mandating high precision and low expression noise such that the appropriate gene expression programs are induced at the correct time and place. If in *Drosophila* patterning mRNAs accumulate in a precise manner minimizing expression noise, as we have shown here, how then can the embryo achieve this near uniformity?

Spatiotemporal Averaging Reconciles Highly Variant Transcription with Precise Accumulation and Recovers the Input-Output Relationship

Large differences in nascent transcript content sustained over sufficiently long periods would inevitably result in unequal transcript production, inconsistent with nearly homogeneous cytoplasmic transcript concentration. As noted above, the long lifetime of *hb* transcripts allows substantial accumulation during the course of the syncytial blastoderm stage. If instantaneous nascent mRNA content is not maintained continuously during

interphase but instead fluctuates about the mean as a result of varying RNAP number, then cytoplasmic accumulation serves as a natural time-averaging filter. The impact of time averaging can be estimated in two independent ways. First, transcript accumulation reflects temporal integration of a signal fluctuating with a characteristic time t_0, the time it takes a polymerase to traverse the 3.2 kbp of the hb gene. RNAP processivity is estimated at 1.1–1.4 kbp/min (Irvine et al., 1991; O'Brien and Lis, 1993; Shermoen and O'Farrell, 1991; Thummel et al., 1990), providing a rough estimate consistent with the observed noise filtering (see Extended Experimental Procedures). Alternatively, a more careful estimate (Figure S6B) yields a theoretical bound on the maximum efficiency of temporal averaging based on directly measured quantities, most crucially, the absolute number of engaged polymerases per nucleus. In the case of *Kr* mRNA profile, by the time the mean expression level reaches 800 molecules per nucleus, pure temporal averaging can at most reduce the expression noise to 8%. For these late embryos, however, our measurements show a consistently lower noise level of 6% ± 2%, suggesting an additional noise-filtering mechanism.

Additional filtering can be readily provided by a small degree of spatial averaging by the exchange of mRNA between neighboring cytoplasmic volumes before the partitioning of the syncytial blastoderm. mRNA possesses some mobility: both *hb* and *Kr* transcript numbers increase at >10 μm from their sites of production in nuclei (Figures S3A and S6A). We note that cylindrical summation volumes with a diameter of one internuclear distance contact each other, so that the mRNA traveling distance required to observe spatial averaging is very small. A straightforward estimate (see Extended Experimental Procedures) shows that attributing the excess noise filtering to spatial averaging requires only 4% of produced transcripts to be exchanged between neighboring volumes. Thus, even a limited degree of spatial averaging is completely sufficient to account for the appearance of low variation in cytoplasmic accumulation from stochastic transcription.

These results have several implications. First, we note that the observed variation in cytoplasmic concentration is likely to contain error introduced by our measurement, and the variation we observe is nearly at the level of counting noise. This might indicate that spatial averaging predominates the filtering of transcription noise; however, the degree to which RNAP numbers fluctuate and, therefore, the extent of purely temporal averaging can only be determined with measurements in living embryos. Second, both spatial and temporal averaging mechanisms effectively relax a requirement for rapid, tightly controlled transcriptional responses to modulating inputs, thereby minimizing the need for additional layers of feedback or other control systems. In turn, the fluctuations of putative inputs must approach the same degree of noise as the intrinsic variability of the transcriptional process itself before any effect on gene expression is realized.

It is well established that the position of the Hb expression boundary depends on Bcd genetic dosage (Driever and Nüsslein-Volhard, 1988) and that the concentration of Hb protein along the AP axis depends upon and is at least as precise as Bcd concentration (Gregor et al., 2007). Superficially, a highly stochastic transcriptional response would appear to render irrelevant any link between Bcd precision and Hb output: the 10% fluctuations observed for Bcd (Gregor et al., 2007) cannot directly impact a transcriptional process whose noise is >40%. However, each nucleus employs averaging mechanisms reducing the effect of intrinsic noise. Because of the central role played by time averaging, the relative importance of various noise sources depends on the timescale of observation. The immediate readout (on a scale of minutes) is dominated by intrinsic transcriptional noise that renders the precision of the input irrelevant. Averaging over active loci, over time, and between neighboring nuclei, the contribution of intrinsic noise becomes comparable with the input (or extrinsic) fluctuations. Thus, on a long timescale, such as 3 hr of development, the precision of patterning decisions becomes limited by the extrinsic variability. A precise response to Bcd will be recovered as long as the mean activity of *hb* transcription is correlated with Bcd concentration, as proposed previously by Erdmann et al. (2009). This reconciles the apparently stochastic behavior of *hb* transcriptional activity with the precisely positioned boundary of expression (Porcher et al., 2010). In this manner, Hb activity and fluctuations in the boundary domain retain the previously observed dependence upon levels of and fluctuations in Bcd concentration.

Limitations to Precise Control of Gene Expression

We have shown that in the context of the gap genes, transcript output in the maximally expressing region does not equate with the actual maximum attainable density of RNAP loading. This maximum is attained by only a small fraction of nuclei at any given moment. Thus, it is currently unclear what determines the mean density of RNAP loading common to these four genes and what prohibits all nuclei from continuously activating the achievable maximum density. It is possible that the output rate is determined by a common, maternally supplied and spatially ubiquitous factor, for example Zelda or BSF (De Renzis et al., 2007; Liang et al., 2008), which calibrates the RNAP density of these four genes to give rise to the observed transcript output rate. Conversely, from the perspective of noise minimization, it would seem advantageous to design these genes' promoter-enhancer architecture such as to obtain the actual maximum possible density because higher output achieves greater noise reduction. However, a biological system likely cannot be readily engineered to produce transcripts at an arbitrarily rapid rate. Hence, it seems likely that mean RNAP loading, and hence transcript output, is strongly influenced by physical considerations, such as transcription factor binding, promoter melting, enhancer looping, and/or chromatin accessibility, that might be difficult to overcome by any simple means. Future work will determine the extent to which the mean polymerase density we observe for these four genes is a shared feature of early expression and the extent to which this rate can be manipulated according to cellular context. Moreover, further studies will be required to determine the timescale of fluctuations of an active locus during interphase, i.e., whether variations arise largely from "bursts" of dense RNAP loading followed by quiescent periods, or conversely, if the variations result from RNAP-loading rates that are maintained continuously during interphase but differ dramatically between loci.

The formation of cellular membranes during the 14th interphase prohibits spatial exchange. It is thus improbable that spatial averaging mechanisms can play a role in ensuring precise responses at this time. Moreover, the transcripts of the pair-rule genes are directed to the apical surface where they accumulate (Davis and Ish-Horowicz, 1991). Differential cellular behavior, presaging the formation of morphological structures, emerges in the latter part of the 14th interphase. Thus, it is likely that shortly after the onset of the 14th interphase, individual cells begin accumulating gene products required for their specific behaviors, thus rendering spatial averaging a hindrance to differentiation. It is therefore likely that temporal averaging and/or other mechanisms such as regulatory feedback ensure the precise distribution of patterning factors at this time. The degree of precision of transcriptional events over the course of the 14th interphase will be the subject of future investigations.

EXPERIMENTAL PROCEDURES

Fly Strains and Embryo Manipulation

Oregon-R (Ore-R) embryos were used as WT. α-Amanitin injection was performed as described by Edgar et al. (1986). RT-qPCR method is described in the Extended Experimental Procedures. Embryos heterozygous for a deficiency spanning *hb* or for the mRNA null Kr^1 mutation were collected from crosses of heterozygous adults (w^{1118}; Df(3R)BSC477/TM6C and Kr^1/SM6) and distinguished from homozygous mutant and homozygous balancer siblings by visual inspection of nascent mRNA sites. Homozygous hb^{12} mutants were obtained from crosses of heterozygous adults carrying a TM3 balancer marked with *hb-lacZ* reporter and identified by the absence of *lacZ* expression. *runt* mutant embryos were collected from crosses of a deficiency-bearing stock (Df(1)BSC645, w^{1118}/Binsinscy) and mutants distinguished by the absence of *runt* expression using FISH probes. gro^{MB36} germline clones were generated using the FLP-FRT recombinase system (Ajuria et al., 2011; Xu and Rubin, 1993). cic^1 homozygous females were crossed to WT males to assay the effect of disabling *capicua* activity.

FISH and Imaging

Embryos were fixed in 5% formaldehyde, 1X PBS for 20 min, and devitellinated as described by Lécuyer et al. (2008). Fixed embryos were rinsed three times in 1X PBS and washed for 10 min in smFISH wash buffer (4X SSC, 35% formamide, 0.1% Tween 20). Hybridization to probes complementary to the reading frame of *hb*, *Kr*, *kni*, or *gt* and conjugated to Atto 565 (Sigma-Aldrich; 72464) or Atto 633 (Sigma-Aldrich; 01464) was performed for 16–24 hr at a concentration of about 1 nM per probe in hybridization buffer (4X SSC, 35% formamide, 10% dextran sulfate, 2 μg/ml BSA [NEB; B9001], 0.1 mg/ml salmon sperm DNA [Invitrogen; 15632-011], and 2 mM ribonucleoside vanadyl complex [NEB; S1402S], 0.1% Tween 20). After two washes of 1 hr in wash buffer, embryos were rinsed twice briefly in 1X PBS, stained with DAPI, and mounted in VECTASHIELD (Vector Laboratories; H-1000). For combined FISH and immunofluorescence, incubation in *hb*-Atto-565 probes was reduced to 2 hr. Hb antibody staining was performed as described by Dubuis et al. (2013) with rat anti-Hb and goat anti-rat Alexa 647. Imaging was performed by laser-scanning confocal microscopy on a Leica SP5 inverted microscope as described (Little et al., 2011) except that we used a 63× HCX PL APO CS 1.4 NA oil immersion objective with pixels of 76 × 76 nm and z spacing of 250 nm. We typically obtained stacks representing 20 μm in total axial thickness starting at the embryo surface. Image analysis was performed as described (Little et al., 2011) with enhancements described in the Extended Experimental Procedures.

SUPPLEMENTAL INFORMATION

Supplemental Information includes Extended Experimental Procedures and six figures and can be found with this article online at http://dx.doi.org/10.1016/j.cell.2013.07.025.

ACKNOWLEDGMENTS

We gratefully acknowledge E. Wieschaus for insightful discussions and continuous support. We thank W. Bialek, S. Blythe, B. Chun, G. Deshpande, S. Di Talia, H. Garcia, S. Kyin, M. Petkova, R. Samanta, T. Schupbach, G. Tkacik, and the Bloomington Drosophila Stock Center. This work was supported by NIH Grants P50 GM071508 and R01 GM097275, and by Searle Scholar Award 10-SSP-274 to T.G.

Received: January 13, 2013
Revised: May 20, 2013
Accepted: July 12, 2013
Published: August 15, 2013

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